Feature-Efficient Machine Learning Framework for Binary and Multiclass Intrusion Detection in IoT Networks

Amr M. El-Husseiny
Integrated Information Systems Center
Information System Department
Cairo, Egypt
amr.elhossieny@egcloud.gov.eg

Ibrahim Bassiony
Department of Computers & A.I
Military Technical College
Cairo, Egypt
i.bassiony@mtc.edu.eg

Ahmed Abdelrahman
Department of Computers & A.I
Military Technical College
Cairo, Egypt
ahmed.abdelrahman@ovgu.com

Ahmed M. Mattar
Department of Computers & A.I
Military Technical College
Cairo, Egypt
a.mattar@ieee.org

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***Abstract*— The rapid growth of Internet of Things (IoT) devices has introduced major security challenges due to the massive volume of network traffic and the resource constraints of IoT environments. Conventional intrusion detection systems (IDSs) and deep learning-based approaches often rely on high-dimensional feature sets and computationally expensive models, which limits their practicality in real-world IoT deployments. This paper proposes FEC-IDS (Feature-Efficient and Complexity-Aware Intrusion Detection System), a lightweight machine learning-based framework for IoT intrusion detection built on systematic feature selection and classical machine learning classifiers. The framework is evaluated using the CICIoT2023 dataset for binary anomaly detection and multiclass signature-based attack classification. Experimental results show that a reduced subset of only seven selected features, combined with Random Forest, achieves 99.76% accuracy in binary detection while significantly reducing training and inference time compared with the full feature set. For multiclass attack category classification, the framework achieves 99.21% accuracy and 96% precision using 16 selected features across eight attack categories. In fine-grained multiclass classification involving 34 attack types, the framework achieves 98.88% accuracy and an F1-score of 75%. Overall, the results demonstrate that FEC-IDS maintains strong detection performance while substantially reducing feature dimensionality and computational cost, making it suitable for deployment in resource-constrained IoT environments. This study highlights the importance of feature efficiency in IoT IDS design and presents a scalable framework that better aligns detection effectiveness with practical deployment requirements.**

***Keywords*— IoT security, intrusion detection system, CICIoT2023, feature selection, anomaly detection, signature-based detection.**

I. INTRODUCTION

The Internet of Things (IoT) underpins contemporary smart environments by interconnecting heterogeneous devices across domains such as smart cities, healthcare, industrial automation, and critical infrastructure. The rapid expansion of IoT ecosystems stems from advances in sensing technologies, wireless communication, and data analytics, enabling real-time collection and processing of large-scale data [1]–[3]. However, this growth also intensifies security challenges, as many IoT devices operate with constraints in processing power, built-in security support, memory, energy, and long operational lifecycles [3], [4].

IoT networks face a broad spectrum of cyber threats. Conventional security mechanisms, including firewalls and rule-based intrusion detection systems (IDSs), often fall short in IoT contexts due to limited effectiveness against evolving or previously unseen attacks. Moreover, centralized security solutions encounter scalability difficulties when handling the substantial and continuous traffic produced by large-scale IoT deployments. Recent surveys indicate that IoT environments remain susceptible to denial-of-service, botnet activity, reconnaissance, spoofing, and protocol exploitation [3]–[7].

To counter these challenges, machine learning–driven intrusion detection approaches have been extensively explored for IoT security. By learning traffic patterns directly from data, these methods can support both anomaly detection and attack classification without sole reliance on manually defined rules [7], [8]. Yet, many existing approaches emphasize detection accuracy at the expense of high-dimensional feature spaces or computationally intensive models. Such designs are difficult to justify in practical IoT settings, where security mechanisms must be lightweight and efficient due to limited processing power, memory, and energy resources [9]–[12].

The availability of realistic, large-scale IoT datasets has accelerated research in this area. Among them, CICIoT2023, introduced by Neto et al. [13], stands out as a comprehensive benchmark for IoT intrusion detection, encompassing a wide array of attack scenarios, categories, and types. Additional datasets such as Bot-IoT, TON_IoT, and IoT-23 have also supported IDS evaluation under realistic IoT traffic conditions [14]–[16]. Recent studies have assessed intrusion detection methods on CICIoT2023 using machine learning, clustering-based, and deep learning approaches, reporting strong results for both binary anomaly detection and multiclass attack classification [17]–[20].

Nevertheless, most studies rely on the full feature set or offer only limited analysis of feature efficiency, increasing computational costs and reducing practicality for deployment on resource-constrained IoT devices [9]–[12].

Motivated by this gap, this work presents FEC-IDS, a feature-efficient and complexity-aware intrusion detection framework for IoT environments. Rather than pursuing higher performance through increasingly complex models or full-dimensional feature sets, the framework prioritizes feature efficiency as a core objective alongside detection effectiveness. Specifically, it investigates whether systematic feature selection can meaningfully reduce model complexity while preserving strong predictive performance across various intrusion detection tasks [12], [21], [22]. The framework accommodates binary anomaly detection and multiclass signature-based attack classification, with careful attention to the trade-off between classification granularity and computational cost.

It is important to note that CICIoT2023 primarily comprises labeled known attack families and types.

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Accordingly, the current evaluation concentrates on discrimination among known traffic classes rather than true zero-day or unknown-attack detection. Extending FEC-IDS to handle novel or previously unseen attacks remains an important direction for future work but lies outside the scope of this study.

Main contributions:

- Introduction of FEC-IDS, a feature-efficient and complexity-aware IDS pipeline for IoT environments, explicitly balancing detection performance with computational overhead rather than prioritizing accuracy alone.
- Development of a task-specific feature selection strategy in which recursive feature elimination (RFE) is applied separately to binary anomaly detection, multiclass attack category classification, and fine-grained attack type classification, yielding compact, task-optimized feature subsets.
- Integration of task-specific RFE with lightweight classical classifiers (Random Forest, Decision Tree, and Logistic Regression), with analysis of the relationship between feature dimensionality, detection performance, and execution cost.
- Comparative evaluation against recent CICIoT2023-based studies, showing that FEC-IDS attains comparable or superior performance while reducing features from 46 to 7, 16, and 30 for the three tasks, respectively, enhancing suitability for resource-constrained IoT deployment.

II. RELATED WORK

The CICIoT2023 dataset, introduced by Neto et al. [13], has significantly influenced contemporary IoT security research. It captures realistic, large-scale IoT network traffic, including benign activity and a diverse set of attacks (denial-of-service, distributed DoS, reconnaissance, brute-force, spoofing, web-based attacks, and botnet-related behaviors). This breadth has made CICIoT2023 a pivotal benchmark for evaluating IoT intrusion detection systems.

Several studies have examined ML-based intrusion detection using CICIoT2023. Kumar et al. [17] evaluated various traditional ML classifiers and found tree-based methods, particularly Decision Tree and Random Forest, to outperform several alternatives in IoT attack detection. Their findings confirm the effectiveness of lightweight classical models but do not explore feature efficiency or how feature dimensionality impacts computational cost and deployment feasibility.

Other studies investigated alternative learning paradigms on CICIoT2023. Gheni and Al-Yaseen [18] proposed a two-step clustering-based intrusion detection approach to address high dimensionality, combining clustering and optimization to achieve strong detection performance. However, these works focus on detection accuracy and robustness, with limited discussion of feature efficiency or resource-constrained deployment.

Recent research has also explored deep learning IDS models on CICIoT2023. Hizal et al. [19] presented a deep learning intrusion detection framework with strong performance across multiple attack scenarios. Alharby et al. [20] evaluated several ML and deep learning models for IoT intrusion detection, reporting competitive results. While effective, deep learning methods generally require greater computational and training resources, which may limit practicality for latency-sensitive, resource-constrained IoT environments.

Consequently, feature selection has become a crucial theme in IoT intrusion detection. Adewole et al. [9] demonstrated that feature selection can reduce dimensionality while improving IDS efficiency. Achbarou et al. [10] showed that effective feature selection can enhance multiclass intrusion detection while lowering computational overhead. Phan et al. [11] proposed lightweight ML-based IDS frameworks for resource-constrained IoT environments and highlighted the need to balance detection effectiveness with model complexity.

Notwithstanding these advances, existing feature selection studies display notable limitations. Many analyses consider a limited number of attack classes or fail to systematically examine the distinct feature requirements of binary anomaly detection, coarse-grained multiclass classification, and fine-grained attack-type classification within a single framework.

In summary, prior work on CICIoT2023 demonstrates that ML, clustering, and deep learning can achieve strong IDS performance. However, most approaches prioritize accuracy with full-dimensional feature sets or computationally intensive models. Few studies systematically investigate feature efficiency across multiple intrusion detection tasks while explicitly analyzing the trade-off between predictive performance and computational cost. This gap motivates the present work, which proposes a lightweight, feature-efficient IDS framework for IoT environments.

III. PROPOSED FRAMEWORK

This section outlines the feature-efficient intrusion detection framework for IoT environments. The framework aims to maintain strong detection performance while reducing computational overhead via task-specific feature selection and lightweight ML classifiers. Fig. 1 illustrates the overall pipeline of the proposed FEC-IDS framework.

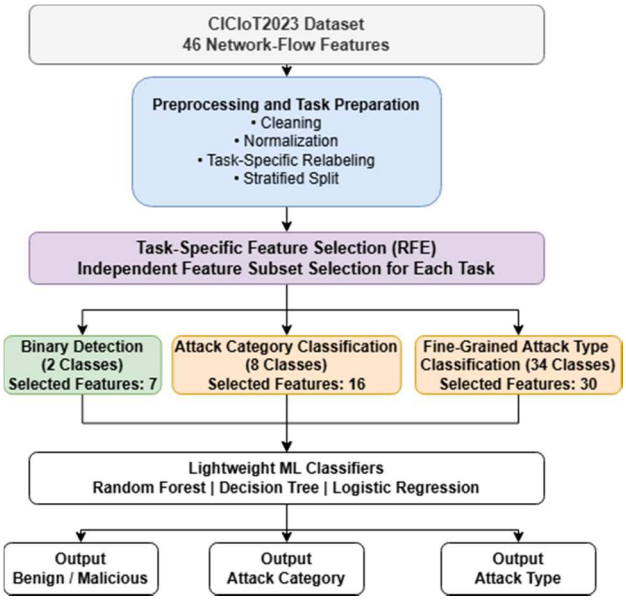


Fig. 1. Overall architecture of the proposed FEC-IDS framework.

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A. Overall Framework Architecture

FEC-IDS is a machine learning–based intrusion detection framework designed for resource-constrained IoT environments. The framework follows a sequential pipeline: dataset preparation and preprocessing, followed by task-specific feature selection with recursive feature elimination (RFE). The resulting feature subsets are used to train lightweight classifiers for three intrusion detection tasks: binary anomaly detection, attack-category classification, and fine-grained attack-type classification.

The framework is suitable for deployment at IoT gateways, edge nodes, or centralized monitoring servers. It supports binary anomaly detection (benign vs. malicious) and multiclass classification (attack categories or specific attack types), offering flexibility for various IoT security scenarios while remaining practical for constrained environments.

B. Data Preprocessing

Data preprocessing is essential for reliable intrusion detection modeling. CICIoT2023 data are processed to improve consistency and enable fair evaluation across tasks.

First, invalid, incomplete, and duplicate records are removed. Numerical features are normalized to reduce scale differences and improve model convergence. The dataset is relabeled according to the target task: binary anomaly detection, attack-category classification, or fine-grained attack type classification. Finally, the data are divided into training and testing subsets using stratified sampling to preserve class distributions across tasks. This preprocessing enhances data quality and ensures consistent, reliable performance comparisons.

C. Feature Selection

Feature selection is central to the framework, balancing accuracy with computational efficiency. Instead of utilizing all features, FEC-IDS applies recursive feature elimination (RFE) to identify compact, informative subsets.

RFE iteratively trains an estimator, ranks features by importance, and removes the least informative features until further removal degrades classification performance. A tree-based estimator is employed within RFE to capture nonlinear relationships and feature interactions common in network traffic. Feature selection is performed independently for each task: binary anomaly detection (fewer features) and multiclass signature-based classification (richer features). This task-specific strategy enables FEC-IDS to balance predictive performance and computational complexity.

D. Classification

Given the resource-constrained IoT context, FEC-IDS relies on lightweight, widely used classifiers: Random Forest (RF), Decision Tree (DT), and Logistic Regression (LR). RF offers robust performance via ensemble learning; DT provides simplicity and fast execution for real-time IoT applications; LR supplies a low-cost linear baseline. Together, these classifiers form a practical basis for assessing how feature reduction affects detection performance and execution efficiency.

Each classifier is trained using both the full feature set and the RFE-selected subsets to evaluate whether model complexity can be reduced without substantial losses in intrusion detection performance.

IV. Experimental Results and Discussion

A. Dataset Description

The framework is evaluated on CICIoT2023, recognized as a comprehensive IoT intrusion detection benchmark [13]. The dataset originates from a large-scale IoT network topology with real devices acting as both targets and sources under realistic conditions.

Table I summarizes benign and malicious traffic distributions. CICIoT2023 comprises 45,588,384 malicious flows across seven attack categories (DDoS, DoS, reconnaissance, web-based attacks, brute-force, spoofing, Mirai botnet activity) and 1,098,195 benign flows. Each record includes 46 statistical traffic features describing protocol behavior and flow dynamics. The dataset supports three intrusion detection tasks through labeling schemes: binary (benign vs. malicious), attack-category, and fine-grained attack-type with 34 distinct types.

TABLE I. TRAFFIC INSTANCE DISTRIBUTION IN THE CICIOT2023 DATASET

Anomaly	Category	Total	Total
Malicious	DDoS	33,984,560	45,588,384
	DoS	8,090,738	
	Recon	354,565	
	Web	24,829	
	Brute Force	13,064	
	Spoofing	486,504	
	Mirai	2,634,124	
Benign	Benign	1,098,195	1,098,195

B. Experimental Setup

The evaluation considers three classification tasks: binary anomaly detection, multiclass attack-category classification, and fine-grained attack-type classification. Experiments run on a workstation with an Intel Xeon Gold 6242R, 3.1 GHz, and 128 GB RAM. Python with standard ML libraries was used.

Fair evaluation employed stratified training/testing splits to preserve class distributions for each task. No cross-validation was used; results reported in Tables II–IV are on held-out test sets. All stochastic operations used a fixed seed for reproducibility.

Default hyperparameters were used for all classifiers, with minor deployment-oriented adjustments. Random Forest used a small ensemble of 10 trees to balance performance and cost; DT and LR followed standard configurations for large-scale tabular data.

Deployment-oriented efficiency is assessed primarily via feature dimensionality, training time, and inference time on the workstation. Memory and energy consumption were not profiled on embedded hardware and are left for future deployment-focused evaluation.

CICIoT2023 is highly imbalanced, with substantially more malicious than benign flows. To reflect natural traffic, original class distributions were preserved with stratified splits. No resampling techniques (e.g., SMOTE) or class-weighting were applied; evaluation used precision, recall, and

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F1-score to reveal potential imbalance effects on detection performance.

C. Evaluation Metrics

Performance is assessed using accuracy, precision, recall, and F1-score, providing a balanced view of classification performance under imbalanced conditions. Metrics are derived from the confusion matrix and applied consistently across experiments for fair comparisons among classifiers and feature configurations.

$$Precision = \frac{TP}{TP+FP} \quad (1)$$

$$Recall = \frac{TP}{TP+FN} \quad (2)$$

$$Accuracy = \frac{TP+TN}{TP+FP+TN+FN} \quad (3)$$

$$F1-score = \frac{2}{\left(\frac{1}{Recall}\right)+\left(\frac{1}{Precision}\right)} \quad (4)$$

D. Binary Anomaly Detection Results

For binary classification (benign vs. malicious), RFE was used with a tree-based estimator, starting from the full 46 features and removing features until a deterioration in performance appeared. The results indicate that a seven-feature subset can achieve performance close to the full feature set while reducing training and inference time substantially.

Fig. 2 illustrates the feature count–performance relationship for the binary task, revealing a saturation point beyond which additional features yield minimal gains.

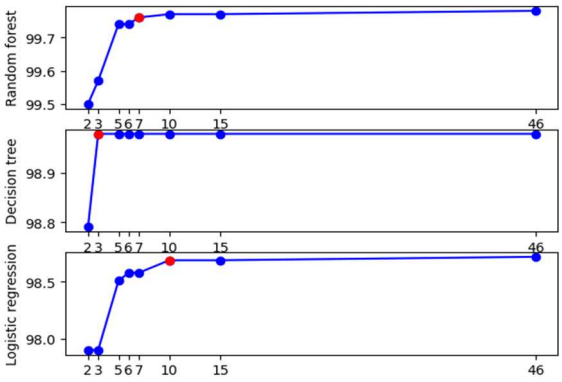


Fig. 2. Performance versus number of features for binary classification.

TABLE II. RESULTS FOR BINARY CLASSIFICATION

Model	Features	Acc	F-score	precision	recall	Train time	Test time
RF	46	99.78	1.00	1.00	1.00	54m3	5.7s
	7	99.76	0.97	0.97	0.98	10m16	3s
DT	46	98.98	0.99	0.99	0.99	6m16	2.0s
	3	98.98	0.90	0.86	0.95	2m12	0.6
LR	46	98.72	0.99	0.99	0.99	7m36	1.1s
	10	98.58	0.81	0.89	0.76	5m18	0.3s

Table II shows full-feature and RFE-subset results for binary anomaly detection. Accuracy and F1-score remain high after feature reduction, and computational costs during training and testing decline markedly. Although the high accuracy and F1-scores reflect the relatively clean separation in CICIoT2023, generalization to other IoT environments should be validated with cross-dataset or cross-environment studies in future work.

E. Interpretation of selected features

The frequently retained features provide insight into discriminative signals for IoT intrusion detection. In binary detection, descriptors such as Magnitude, Radius, Variance, Weight, Header_Length, and rst_count capture general deviations in traffic intensity, flow regularity, and packet structure distinguishing benign from malicious communications. Benign IoT traffic tends to be periodic and stable, whereas attacks such as DDoS, scanning, and brute-force generate abrupt changes in packet volume, header behavior, and reset activity, reflected in these features.

For the 8-class and 34-class tasks, the selected subsets also include protocol-level and flag-level indicators (e.g., Protocol Type, UDP, HTTP, HTTPS, ack_count, syn_flag_number, psh_flag_number, urg_count, fin_count, ack_flag_number), along with temporal and rate descriptors (Duration, flow_duration, IAT, Rate, Srate) and higher-order statistics (Max, Min, AVG, Std, Tot_sum, Tot_size, Number, Covariance). Collectively, these features capture how attack families manipulate transport-layer behavior, application-layer usage, timing patterns, and flow burstiness, which are particularly discriminative when normal IoT traffic is regular.

F. Multiclass Attack Classification Results

Two multiclass scenarios are evaluated: (i) attack-category classification with eight classes and (ii) fine-grained attack-type classification with 34 classes. RFE is applied independently to identify task-specific feature subsets that balance performance and efficiency.

a) Eight-Class Attack Category: Classification The dataset is partitioned into eight classes (seven malicious categories plus benign). This task demands more features than binary detection to distinguish among several attack groups. The optimal RFE subset comprises 16 features; further reduction degrades performance.

Table III presents full-feature and RFE-subset results for eight-class classification, showing comparable accuracy and F1-score with reduced training and inference time. This demonstrates that a substantial portion of the original feature space can be pruned without major loss in category discrimination. Fig. 3 shows a saturation pattern: performance gains plateau as features accumulate.

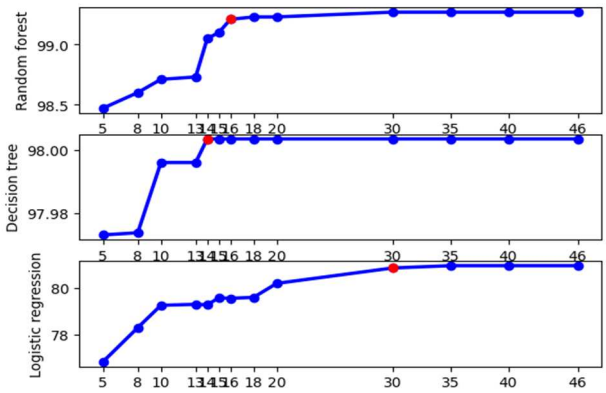


Fig. 3. Performance versus number of features for 8-class classification.

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TABLE III. RESULTS FOR 8-CLASS ATTACK CATEGORY CLASSIFICATION

Model	Features	Acc	F-score	precision	recall	Train time	Test time
DT	46	98.0035	0.51	0.49	0.53	4m38	1.8s
	14	98.0035	0.51	0.49	0.53	2m11	1.2s
RF	46	99.27	0.88	0.96	0.82	130m21	11.4s
	16	99.21	0.87	0.96	0.81	44m43	8.4s
LR	46	80.95	0.44	0.52	0.43	44m10	1.8s
	30	80.85	0.45	0.51	0.44	40m9	1s

b) Thirty-Four-Class Fine-Grained Attack Classification : For 34 attack types, the model must resolve subtle differences among behaviors, making this the most challenging task. The RFE analysis indicates an optimal subset of 30 features. Reducing below this level substantially harms performance, especially for tree-based models.

Table IV reports results for fine-grained classification with full vs. reduced features. The reduced 30-feature subset preserves the same overall accuracy and F1-score for Random Forest while lowering training and inference time. This underscores the practical benefits of feature reduction even in highly granular intrusion detection tasks. The larger gap observed for Logistic Regression is expected due to linear boundaries being less suited to nonlinear, overlapping distributions across 34 attack types. Fig. 4 displays the feature-count vs. performance relationship for the 34-class task, confirming a saturation region similar to prior experiments.

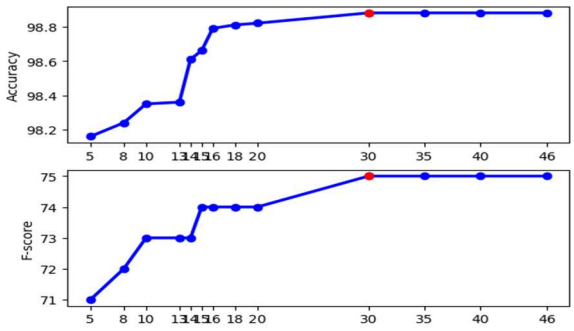


Fig. 4. Performance versus number of features for 34-class classification.

TABLE IV. RESULTS FOR 34-CLASS FINE-GRAINED ATTACK CLASSIFICATION

Model	Features	Acc	F-score	precision	recall	Train time	Test time
DT	46 features	50.10	0.12	0.14	0.14	5m15	3.4s
RF	46 features	98.88	0.75	0.75	0.84	60m6	10s
	Best 30 features	98.88	0.75	0.75	0.84	50m20	8.5s
LR	46 features	78.61	0.41	0.45	0.42	155m18	5.9s

G. Ablation Study and Feature Efficiency Analysis

An ablation study across tasks reveals similar saturation behavior: rapid improvements when top features are retained, followed by stabilization as additional features contribute limited value (Figs. 2–4). Figure 5 shows training time increasing with feature count; Random Forest incurs higher costs than DT and LR due to ensemble operations, though feature reduction mitigates this overhead. The 10-tree RF configuration was chosen to reflect deployment-oriented constraints.

The ablation analysis confirms that systematic feature selection is crucial for balancing detection accuracy and computational efficiency in IoT intrusion detection. Utilizing substantially smaller feature subsets reduces training and inference costs and decreases the feature-vector size required for storage and processing, supporting practical deployment on resource-constrained IoT gateways and edge nodes.

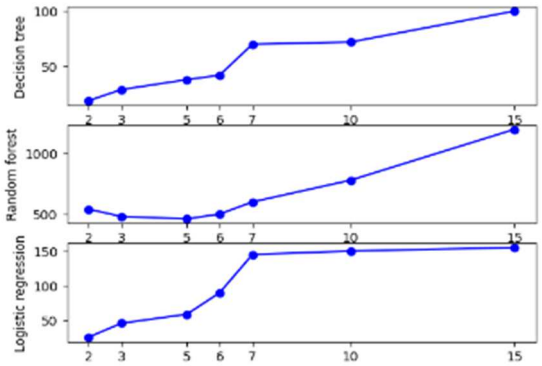


Fig. 5. Training time versus number of features.

Overall, consistently strong F1-scores across binary and multiclass tasks suggest the selected feature subsets remain effective despite CICIoT2023's class imbalance, even without explicit resampling.

H. Comparative Analysis with Existing Work

Table V compares FEC-IDS with recent CICIoT2023-based studies. The proposed framework achieves competitive or superior performance while using substantially fewer features.

In binary anomaly detection, FEC-IDS attains strong performance with only seven features, in contrast to 46 features in prior work. Feature dimensionality is similarly reduced for the 8-class and 34-class scenarios without substantial performance loss. These findings highlight the framework's primary strength: feature efficiency.

Whereas prior studies mainly emphasized accuracy, FEC-IDS explicitly addresses the trade-off between predictive performance and computational cost, enhancing practicality for IoT environments with limited processing power, memory, and energy resources.

TABLE V. COMPARATIVE RESULTS.

Classification	Research	features	Accuracy	precision	recall	F-score
Binary category	FEC-IDS	7	99.76	0.97	0.98	0.97
	Study [17]	46	99.68	0.96	0.96	0.96
	Study [18]	46	99.26	0.99	0.99	0.99
Class 8 category	FEC-IDS	16	99.21	0.96	0.81	0.87
	Study [17]	46	99.40	0.91	0.718	0.705
	Study [18]	46	99.40	0.910	0.710	0.705
Class 34 category	FEC-IDS	30	98.88	0.75	0.84	0.75
	Study [17]	46	99.10	0.80	0.715	0.7064
	Study [18]	46	99.10	0.831	0.714	0.704

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V. CONCLUSION

This paper presented FEC-IDS (Feature-Efficient and Complexity-Aware Intrusion Detection System), a lightweight and feature-efficient intrusion detection framework for IoT environments that explicitly balances detection effectiveness with computational complexity. The framework was evaluated on the CICIoT2023 dataset across three intrusion detection tasks with increasing classification granularity.

The experimental results showed that binary anomaly detection can be performed effectively using only seven selected features, substantially reducing feature dimensionality while maintaining strong detection performance. FEC-IDS also achieved strong performance in multiclass attack-category classification using 16 features across eight classes. In the most challenging setting, fine-grained attack-type classification over 34 classes, an optimized subset of 30 features achieved a practical balance between detection accuracy and computational cost. Across all tasks, Random Forest consistently outperformed Decision Tree and Logistic Regression.

Although Random Forest is inherently more computationally demanding because of its ensemble structure, the feature reduction strategy used in FEC-IDS significantly reduced this overhead and led to shorter training and inference times. The ablation analysis further confirmed that systematic feature selection is an effective mechanism for controlling model complexity without causing a substantial loss in detection performance.

A comparison with recent studies based on the same dataset showed that FEC-IDS achieves comparable, and in some cases superior, detection performance while using substantially fewer features and lower computational cost. These findings highlight the importance of task-specific feature selection in enabling practical intrusion detection for resource-constrained IoT environments.

Nevertheless, the current assessment is limited to the CICIoT2023 benchmark and does not establish cross-dataset generalization or robustness against previously unseen attack behaviors beyond the labelled signatures included in this dataset. Future work will therefore investigate cross-dataset validation, the detection of unknown or evolving IoT attacks, adaptive feature selection strategies, online learning mechanisms, and deployment-oriented evaluation on representative IoT gateways and edge devices with explicit measurement of memory usage, latency, and energy consumption.

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